

Baroclinic critical layers and zombie vortices

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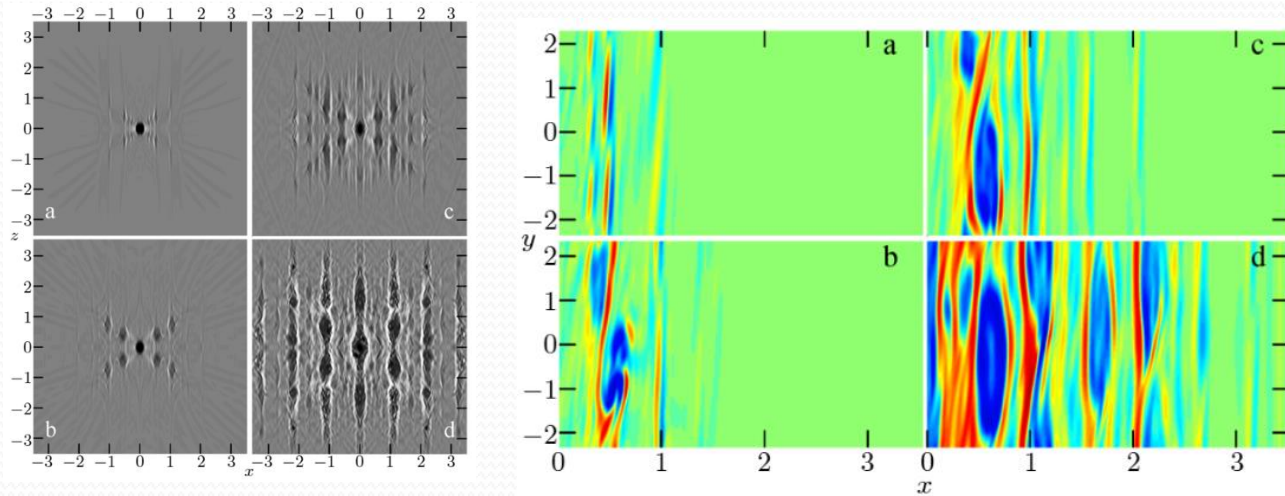
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The zombie vortex instability

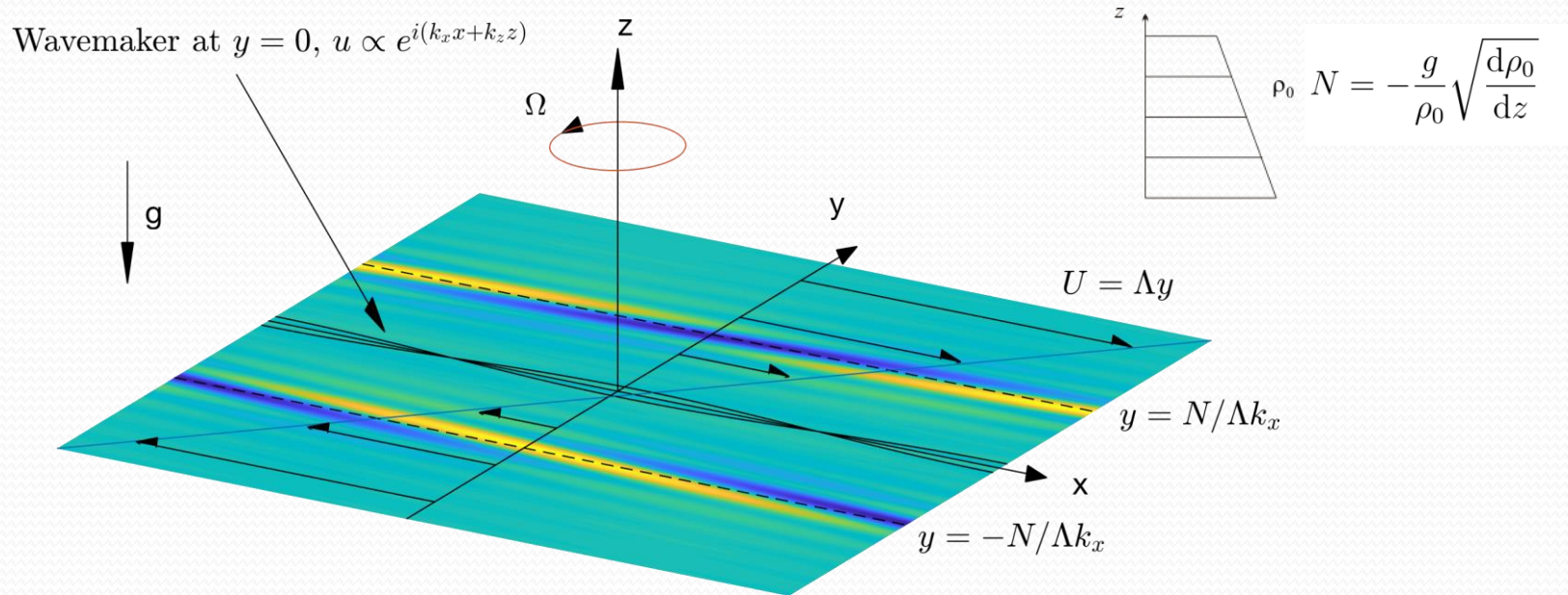
- Self-replicating vortices (Marcus *et al.* 2013)



- Excitation of baroclinic critical layers
- Potentially important in protoplanetary disks
- We undertake a theoretical analysis to understand it
 - Analytical solutions / reduced models

The model

- Horizontal shear, density stratification, planetary rotation
- Steady forcing



$$\mathcal{N} = \frac{N}{\Lambda} \quad f = \frac{2\Omega}{\Lambda} \quad m = \frac{k_z}{k_x}$$

Governing equations

- Fluid equations with Boussinesq approximation

$$\begin{aligned}u_t + yu_x + (1 - f)v + uu_x + vu_y + wu_z &= -p_x + \nu \nabla^2 u \\v_t + yv_x + fu + uv_x + vv_y + wv_z &= -p_y + \nu \nabla^2 v \\w_t + yw_x + uw_x + vw_y + ww_z &= -p_z - \rho + \nu \nabla^2 w \\\rho_t + y\rho_x - \mathcal{N}^2 w + u\rho_x + v\rho_y + w\rho_z &= \lambda \nabla^2 \rho \\u_x + v_y + w_z &= 0\end{aligned}$$

- Initial and boundary conditions

$$u = v = w = \rho = p = 0 \quad t = 0$$

$$u|_{y=0+} - u|_{y=0-} = \varepsilon_0 \exp(ix + imz) + \text{c.c.} \quad t > 0$$

The steady wave solution

- The steady forcing drives linear steady waves in ideal shear flow

$$(u, v, w, p, \rho) = [\hat{u}(y), \hat{v}(y), \hat{w}(y), \hat{p}(y), \hat{\rho}(y)] \exp(ix + imz) + \text{c.c.} \quad (c = 0)$$

$$\hat{p}'' - \frac{2y}{y^2 - f(f-1)} \hat{p}' - \left[\frac{y^2 - f(f+1)}{y^2 - f(f-1)} + m^2 \frac{y^2 - f(f-1)}{y^2 - \mathcal{N}^2} \right] \hat{p} = 0$$

- Singularities arise at $y = \pm \mathcal{N}$

- Baroclinic critical levels

$$U(y) - c = \pm \frac{N}{k}$$

- The steady wave solution diverges

$$\hat{p} \rightarrow A, \quad |A| = \varepsilon$$

$$\hat{u}, \hat{v} \sim \log(y - \mathcal{N})$$

$$\hat{w}, \hat{\rho} \sim \frac{1}{y - \mathcal{N}}$$

The linear critical layer

- The flow in the critical layer is unsteady

$$(u, v, w, p, \rho) = [\tilde{u}(y, t), \tilde{v}(y, t), \tilde{w}(y, t), A, \tilde{\rho}(y, t)] \exp(ix + imz) + \text{c.c.}$$

- Local analysis

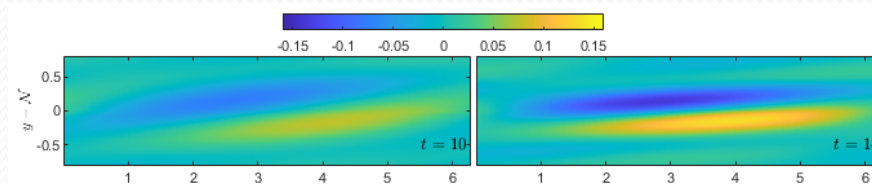
$$\tilde{w}_t + iy\tilde{w} + \tilde{\rho} = -imA$$

$$y \approx \mathcal{N}, \quad t \gg 1$$

$$\tilde{\rho} + iy\tilde{\rho} - \mathcal{N}^2\tilde{w} = 0$$

$$\tilde{\rho} = -\frac{i}{\mathcal{N}}\tilde{w} = -\frac{1}{2}im\mathcal{N}A \frac{e^{-i(y-\mathcal{N})t} - 1}{y - \mathcal{N}}$$

- The wave amplitude increases $\tilde{\rho}, \tilde{u}_y \sim \varepsilon t$
- The critical-layer thickness decreases $|y - \mathcal{N}| \sim t^{-1}$



Evolution of vorticity $v_x - u_y$

Stewartson, K. 1978. The evolution of the critical layer of a Rossby wave. *Geophys. Astrophys. Fluid Dyn.* 9:185–200.

Warn, T. and Warn, H. 1978. The evolution of a nonlinear critical level. *Stud. Appl. Math.* 59:37–71.

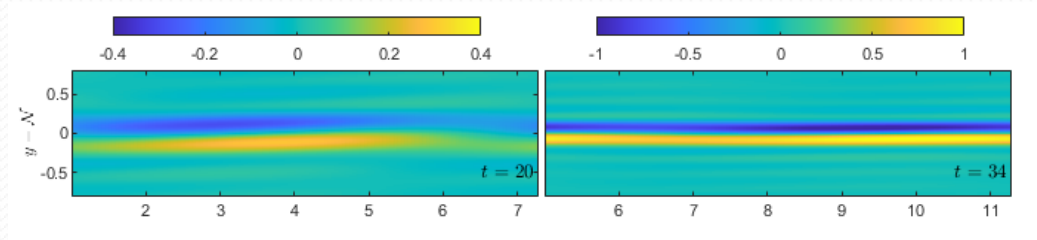
The mean-flow response

- Growing waves drive zonal mean flow

$$(u, v, w, p, \rho) = [\tilde{u}(y, t), \tilde{v}(y, t), \tilde{w}(y, t), A, \tilde{\rho}(y, t)] \exp(ix + imz) + \text{c.c.} \\ + [u_0(y, t), v_0(y, t), w_0(y, t), p_0(y, t), \rho_0(y, t)]$$

$$\frac{\partial u_0}{\partial t} = -\tilde{v}^* \tilde{u}_y + im \tilde{w} \tilde{u}^* + \text{c.c.}$$

$$u_0 = \frac{m^2 |A|^2}{\mathcal{N}} \frac{\cos t(y - \mathcal{N})}{(y - \mathcal{N})} \sim \varepsilon^2 t^2$$



Evolution of vorticity $v_x - u_y$

- Harmonics remain weak because they have different critical levels

$$U(y) - c = \pm \frac{N}{k}$$

The secondary instability

- Mean-flow defect induces instability (cf. Umurhan et al. 2016)

$$u_0 = \frac{m^2 |A|^2}{\mathcal{N}} \frac{\cos t(y - \mathcal{N})}{(y - \mathcal{N})} \sim \varepsilon^2 t^2$$

- Instability happens when

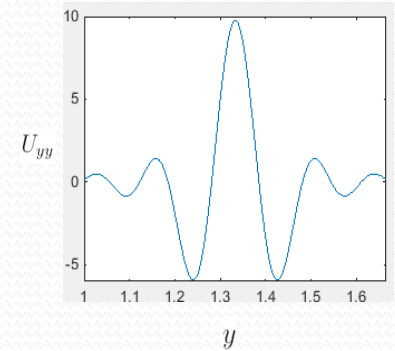
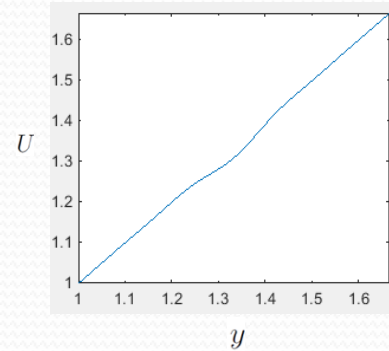
$$u_{0,yy} \sim O(1) \quad \Rightarrow \quad t \sim O(\varepsilon^{-\frac{1}{2}})$$

- A reduced model

$$t = \varepsilon^{-\frac{1}{2}} T, \quad y - \mathcal{N} = \varepsilon^{\frac{1}{2}} Y, \quad u_0 = \varepsilon \mathcal{U}, \quad \nu = \varepsilon^{\frac{3}{2}} \mu \quad \theta = x + mz - \mathcal{N}t$$

$$\mathcal{Z}_T + Y \mathcal{Z}_\theta + \Phi_\theta (\mathcal{Z}_Y - \mathcal{U}_{YY}) = \mu \mathcal{Z}_{YY}$$

$$\int_{-\infty}^{\infty} \mathcal{Z} dY = \sum_{n=1}^{\infty} n K \Phi_n \frac{1-f}{f} (\alpha_+ - \alpha_-) e^{inK\theta} + \text{c.c.}$$



Linear instability

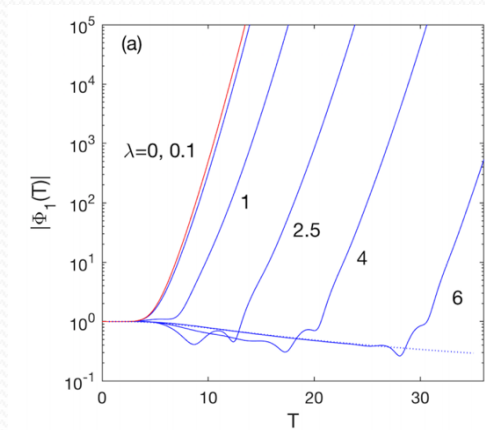
- Linearised equation

$$\mathcal{Z}_T + Y \mathcal{Z}_\theta + \Phi_\theta (\mathcal{Z}_Y - \mathcal{U}_{YY}) = \mu \mathcal{Z}_{YY}$$

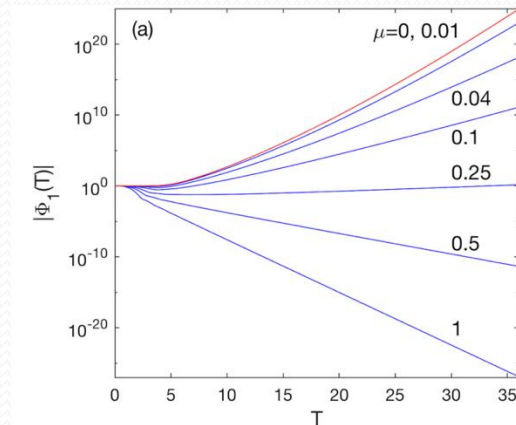
- Ideal fluid ($\mu = \lambda = 0$)

$$\Phi \sim \frac{a}{T^{K+\frac{1}{3}}} \exp \left\{ \frac{3}{8} [\sqrt{3} - i \operatorname{sgn}(\sigma)] \sigma^{\frac{1}{3}} T^{\frac{4}{3}} + i K \theta \right\} \quad \left(\mathcal{U} = \frac{m^2 \cos YT - 1}{\mathcal{N} Y^2} \right)$$

- Effects of dissipation



Thermal diffusivity



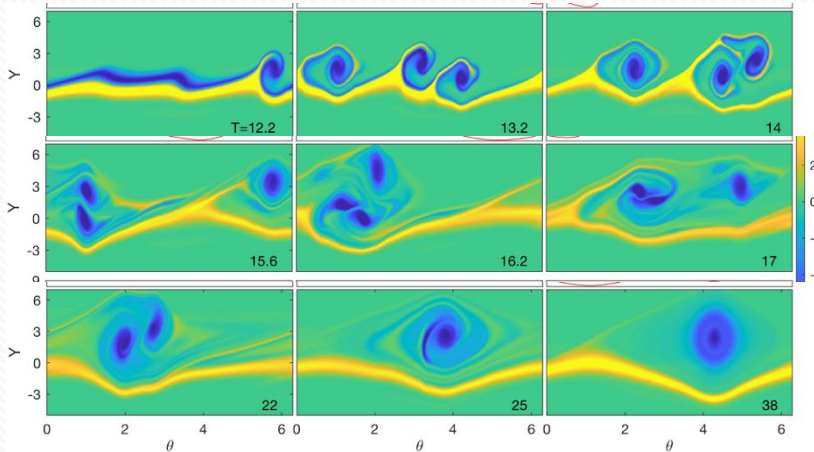
Viscosity

Nonlinear evolution

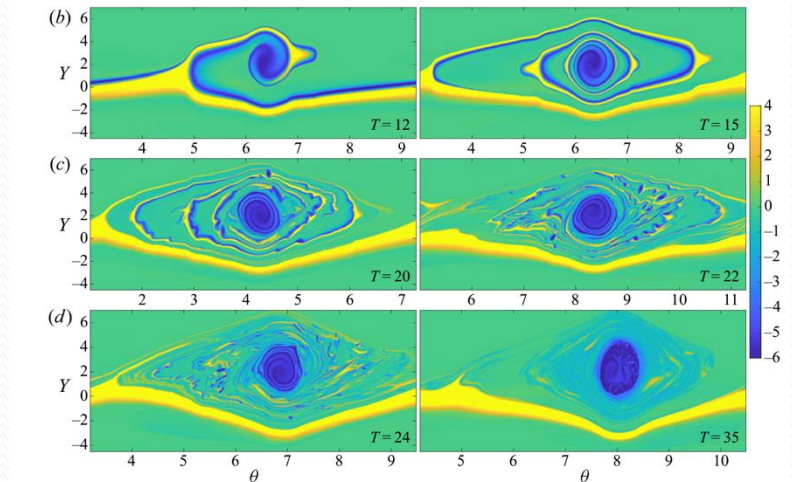
- Numerical solution of the reduced model

$$\mathcal{Z}_T + Y \mathcal{Z}_\theta + \Phi_\theta(\mathcal{Z}_Y - \mathcal{U}_{YY}) = \mu \mathcal{Z}_{YY} \quad \int_{-\infty}^{\infty} \mathcal{Z} dY = \sum_{n=1}^{\infty} n K \Phi_n \frac{1-f}{f} \alpha e^{inK\theta} + \text{c.c.}$$

- Rollup into vortices



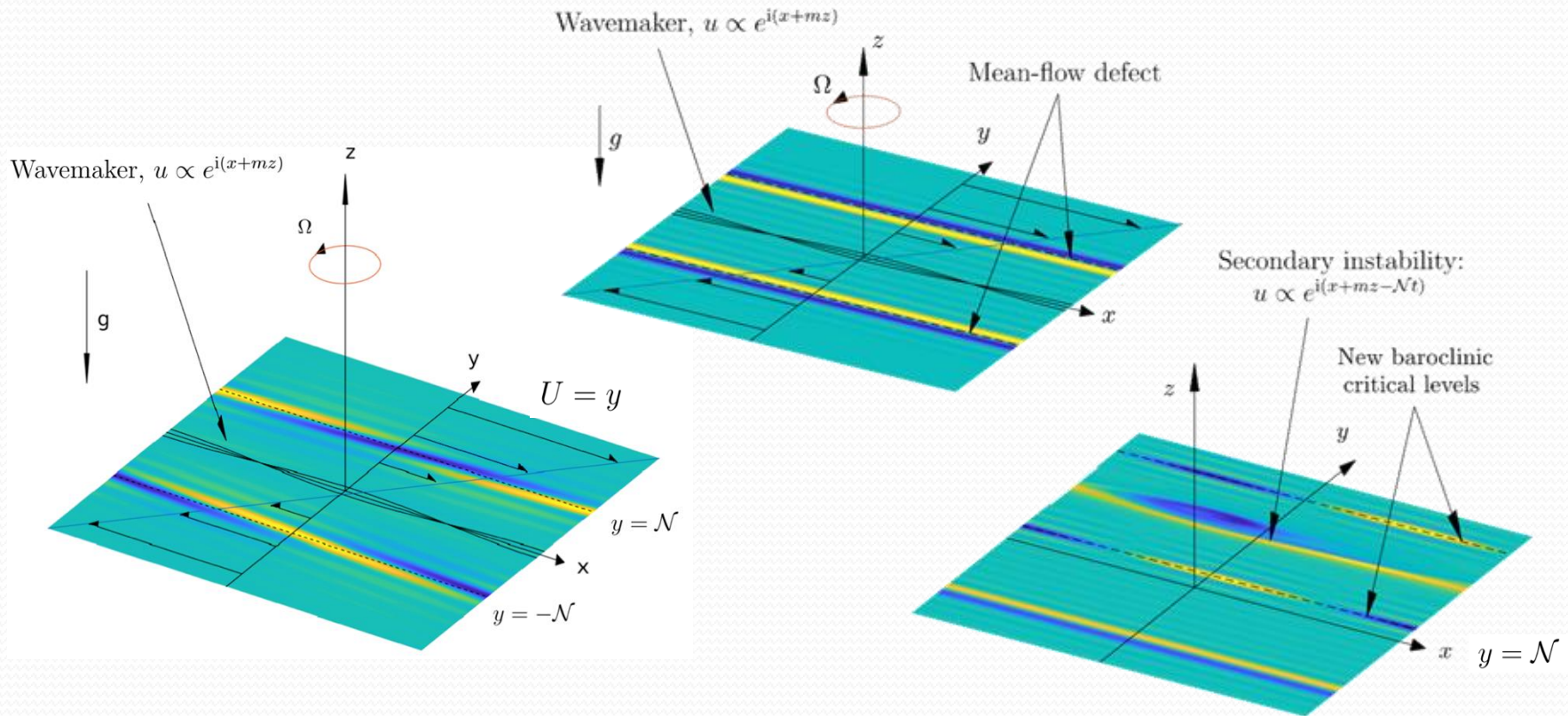
Initial condition of Gaussian distribution, $\mu = 0.025$.



Initial condition of a Fourier mode, $\mu = 10^{-4}$

Reiplication

- The secondary instability can excite new critical layers



Wang, C. and Balmforth, N.J., 2020. Nonlinear dynamics of forced baroclinic critical layers. *J. Fluid Mech.*, 883, A12.

Wang, C. and Balmforth, N.J., 2021. Nonlinear dynamics of forced baroclinic critical layers II. *J. Fluid Mech.*, 917, p.A48.

Relevance to accretion disks

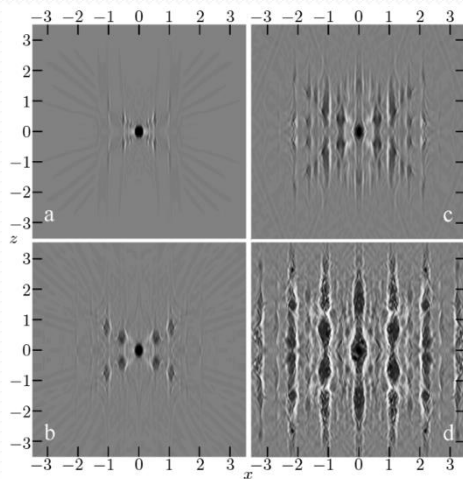
- Zombie vortex instability vs traditional linear instability
- Pros: takes place for wider range of flows and parameters, e.g. linearly stable flows, dead zones
- Cons: more demanding on the initial disturbance
 - The forcing that serves at the initial vortex is

$$u|_{y=0+} - u|_{y=0-} = \varepsilon_0 \exp(ix + imz) + \text{c.c.}$$

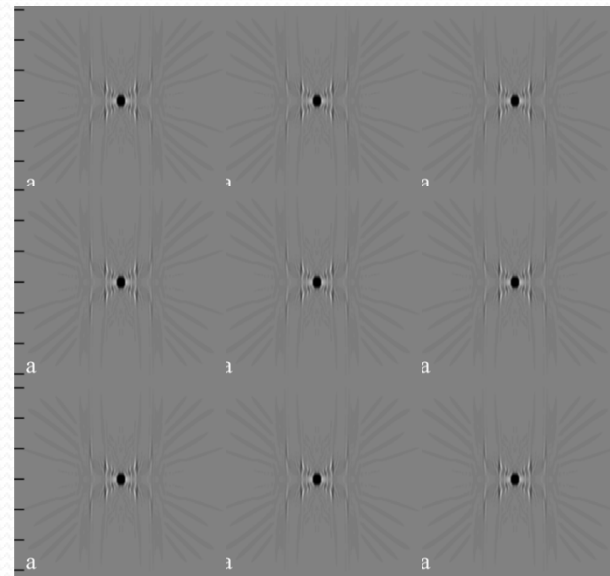
- It is periodic in both x and z directions, which is crucial for baroclinic critical layers.
- What if the initial disturbance is not like this?

Full numerical simulations

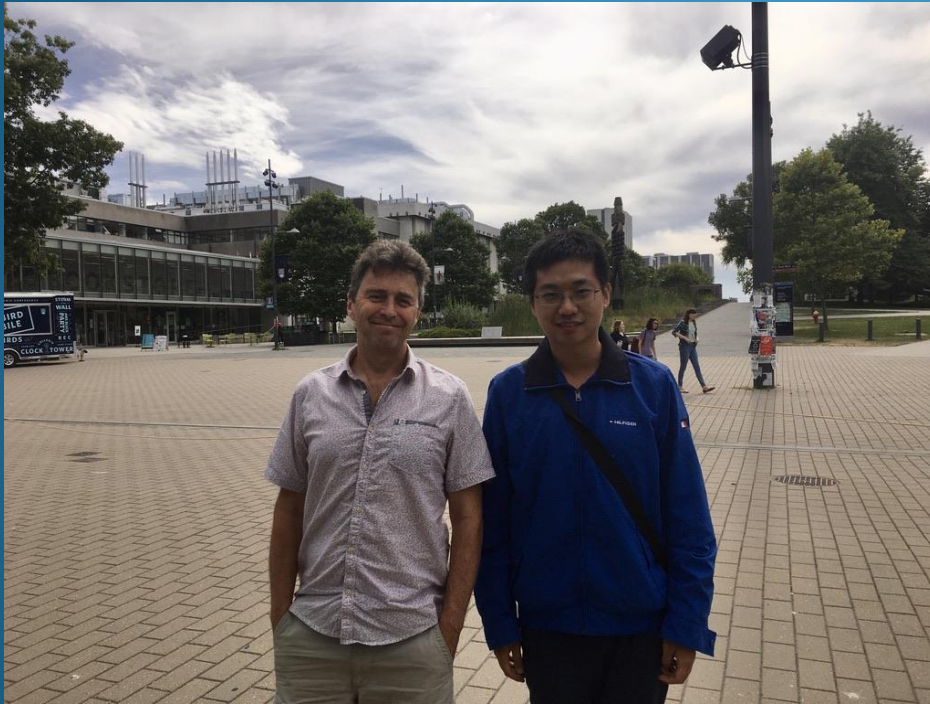
- Do periodic boxes bring in additional (or spurious?) periodicity?



Marcus et al (2013)



The actual initial condition



Thank you